

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech. Summer 2017 - 18 Examination

Semester: 5
Subject Code: 03105303
Subject Name: Theory of Computation

Date: 02/06/2018
Time: 10:30 am to 1:00 pm
Total Marks: 60

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions - (All are compulsory) (Each of one mark) (15)

1. Minimum length of the string formed by regular expression $(a+b)^* aba^* (a+b)^*$
- A) 2 B) 6 C) 4 D) 8

2. Which of the following FA produces the output

- A) DFA
B) NFA
C) Moore Machine
D) None of the above

3. $S \rightarrow aSa \mid bSb \mid a \mid b$

The language generated by the above grammar over the alphabet $\{a, b\}$ is the set of.

- A) All palindromes
B) All odd length palindromes
C) String that begin and end with same symbol.
D) All even length palindromes

4. Which one of the following language over the alphabet $\{0, 1\}$ is described by the regular expression: $(0 + 1)^* 0 (0 + 1)^* 0 (0 + 1)^*$?

- A) The all set of strings containing the substring 00.
B) The set of all strings containing at most two 0's.
C) The set of all strings containing at least two 0's.
D) The set of all strings that begin and end with either 0 or 1.

5. A Context free grammar is

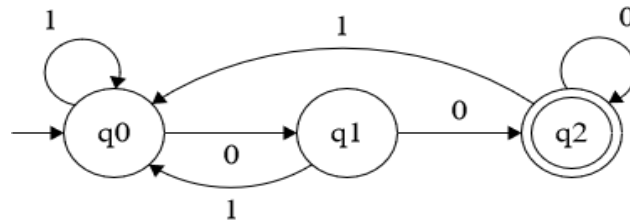
- A) Type-0 B) Type-1 C) Type-2 D) Type-3

6. The language accepted by a Push down Automata is _____ language.

7. There are _____ tuples in finite state machine.

- A) 4
B) 5
C) 6
D) unlimited

8.



The above DFA accepts the set of all strings over $\{0,1\}$ that

- A) Begin either with 0 or 1.
- B) End with 0.
- C) Ends with 00.
- D) Contains the substring 00.

9. Transition function of DFA maps $\delta =$ _____

10. Minimum numbers of states required to construct DFA which accepts all the strings of 0's and 1's where every string starts with '01' is _____.

11. Write the condition for the left recursive grammar.

12. A regular language could have strings of infinite length True/False

13. The union of two context free language is context free. True / False

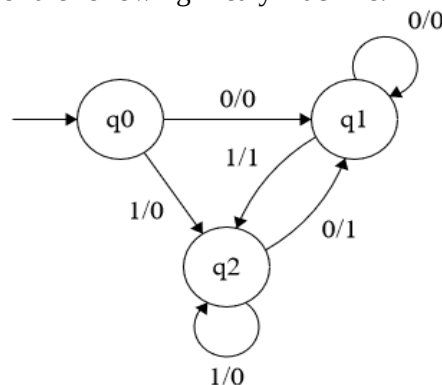
14. Give the name of undecidable problem for CFG.

15. Define Finite Language.

Q.2 Answer the following questions. (Attempt any three)

A) Construct Moore machine for the following Mealy machine.

(05)



B) 1) Explain the types of Turing machine.

(03)

2) Define ambiguity of Grammar.

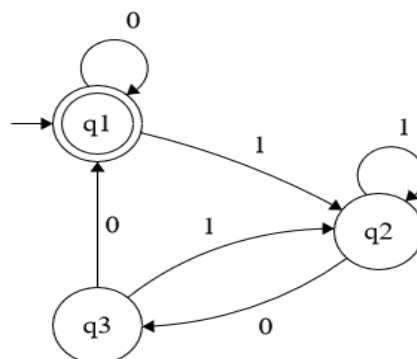
(02)

C) Write an explain Myhill-Nerode theorem.

(05)

D) Define ARDEN's Theorem. Construct a regular expression corresponding to the automata given below using ARDEN's Theorem.

(05)



Q.3 A) What do you mean by Normal forms? Reduce the grammar G with the following productions to CNF. (07)

$$S \rightarrow ASB$$

$$A \rightarrow aAS \mid a \mid \varepsilon$$

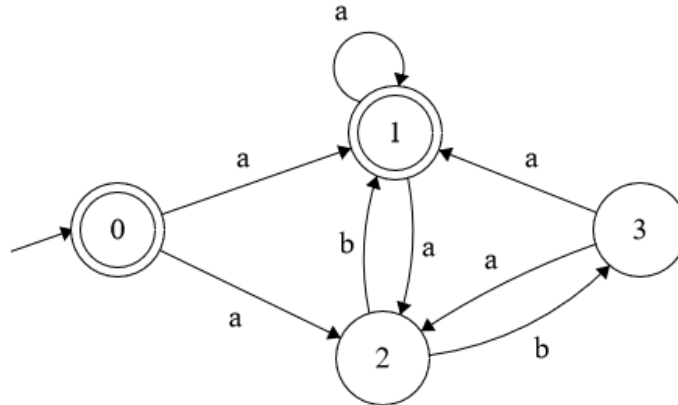
$$B \rightarrow SbS \mid A \mid bb$$

B) Define Turing machine and Design a Turing Machine for the following Language: (08)

$$L = \{a^n b^n \mid n \geq 0\}$$

OR

B) 1) Construct DFA equivalent to given NFA. (04)



2) Define pumping lemma for regular language. Prove by pumping lemma the following language (04)

$$L = \{a^i b^i \mid i \geq 0\} \text{ is not regular.}$$

Q.4 A) Define PDA also Construct a PDA for the following Language: (07)

$$L = \{a^n b^m c^n \mid m, n \geq 0\}$$

OR

A) 1) For the following Regular Expression draw a DFA recognizing the corresponding language. (04)

$$(0 + 1)^* 10 (0+1)^*$$

2) Explain recursively enumerable language. (03)

B) 1) Draw Deterministic Finite Automata for the following languages (04)

i) $L_1 = \{x \in (0,1)^* \mid x \text{ contains } 101 \text{ as a substring}\}$

ii) $L_2 = \{x \in (0,1)^* \mid x \text{ contains odd number of zero}\}$

2) Using principle of Mathematical induction, Prove that for every $n \geq 1$, (04)

$$1 + 3 + 5 + \dots + (2n - 1) = n^2$$